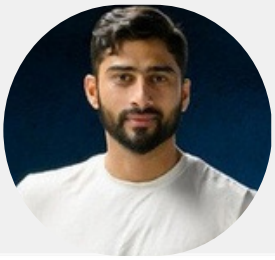
MUHAMMAD SHAHZAIB

Environmental professional with 4+ years of experience supporting industry and organizations in achieving compliance and sustainability objectives through applied expertise and practical technical solutions.



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Helsinki, Finland

EDUCATION

Master's in Environmental Technology

University of Eastern Finland (2023 – 2025)

Bachelor's in Environmental Science

University of the Punjab (2016 – 2020)

SKILLS & KNOWLEDGE

- **Environmental Compliance & Regulation:** EIA, EMS (ISO 14001), QMS (ISO 9001), EHS, environmental audits, Regulatory compliance and permitting, environmental audits and reporting (incl. third-party audits)
- **Environmental Monitoring:** Air quality monitoring, water quality monitoring, waste management monitoring, emissions measurement, GHG accounting, industrial and site environmental monitoring, Analytical instrumentation (measurement methods, devices, laboratory-based analysis tools), instrumentation calibration, troubleshooting & validation
- **Data Analysis & Reporting:** ArcGIS (environmental/spatial analysis-basic), OneClick LCA, environmental data analysis, real-time monitoring, PowerPoint reporting systems & dashboards, Excel, MATLAB, Python (Numpy, Pandas)
- **Sustainability & Environmental Systems:** Waste management, circular economy, construction environmental compliance, process optimization based on environmental performance data, environmental reporting, and technical documentation
- **Project Management & Communication:** Cross-functional collaboration (engineering, R&D, sustainability teams), regulatory stakeholder coordination, project monitoring and evaluation, technical communication, and reporting

TRAININGS & CERTIFICATIONS

- ESG (reporting, certifications, and regulatory frameworks - The ESG institute)
- Digital tools and Data Analytics (UEF)
- Occupational Safety (Kiwa Suomi)
- Embodied Carbon, LCA, EPD (Paroc Oy)
- Environmental Monitoring and Risk Management (UEF)
- Project management and leadership (Talent Hub Eastern Finland
- Environmental law and regulations (University of the Punjab)
- Industrial Safety Management (University of the Punjab)
- Professional competence certificate (Entrepreneurship and Sustainability, YUFE)
- Carbon Dioxide Removal - CDR (NASDAQ)

PROFESSIONAL EXPERIENCE

Environmental Intern | Environmental Management

Paroc, Finland | 2025- Present

Manufacturer of stone wool insulation solutions for energy-efficient and sustainable construction

- Planned and executed **10+ emissions measurement campaigns** across multiple process units within a manufacturing environment, ensuring alignment with EU ETS requirements, BAT-associated emission limits, and relevant **ISO standards**.
- Performed on-site **emissions monitoring** using both online and offline instrumentation, integrating process data from operational systems with **laboratory analysis** of collected samples.
- Analyzed emissions and process datasets to identify root causes of deviations, supporting targeted recommendations for process improvements and **regulatory compliance**.
- Improved GHG inventory accuracy by implementing structured **QA/QC** procedures and conducting systematic validation of emissions data to enhance reliability and traceability.
- Developed and applied in-house measurement and calculation methodologies aligned with **official reporting standards**, strengthening emissions profiling and reporting consistency.
- Produced technical **environmental and compliance reports** to support regulatory submissions and internal sustainability decision-making.
- Conducted environmental compliance assessments against **EU regulatory frameworks**, identifying gaps and supporting actions to address compliance risks.
- Coordinated findings with senior management, contributing to **gap analysis** and **process optimization** initiatives to improve environmental performance.
- Supported third-party environmental compliance audits by preparing data, performing detailed checks, and collaborating with auditors to clarify results and support follow-up actions.
- Collaborated with **R&D, Sustainability**, and **Circular Economy** teams to evaluate environmental performance of ongoing projects and inform planning for upcoming developments.

Research Assistant | Environmental & Emissions Research

University of Eastern Finland | 2023 - 2024

FINE Aerosol Research Group (ILMARI Lab) – aerosol and emissions research

- Contributed to EU-funded **environmental research projects**, supporting experimental design, emissions measurement campaigns, and technical reporting in collaboration with international partners.
- Build a small **environmental modeling** project using online Hyytiälä monitoring station data (SMEAR II) to assess the seasonal and yearly emission trends in the Helsinki region.
- Supported the **design** and **setup of experimental systems**, including emissions sampling lines, filter collection systems, and measurement configurations.
- Performed **calibration, testing**, and **troubleshooting** of measurement instruments, ensuring data accuracy and reliability.
- Conducted laboratory-based emissions measurements and sample collection, including filter-based sampling for subsequent analysis.
- Performed **data analysis**, with a focus on physicochemical properties, assessing air quality impacts and environmental performance under varying operating conditions.
- Translated complex experimental data into **technical reports**, **scientific publications**, and **presentations**, contributing to dissemination and group-level discussions.
- **Co-authored** conference papers and a peer-reviewed publication, demonstrating strong technical documentation and communication skills.

VOLUNTEERING

- Finnish Red Cross (**Punainen Risti**): Volunteering for charity or donation collection, organizing training and workshops
- Student Tutor at the University of Eastern Finland: to support students in their settlements and head start in their education
- JCI (**Junior Chamber International**) Finland: Assisted in organizing events and workshops, and combined engagement sessions with young people

LANGUAGES

- Urdu (*Native*)
- **English** (*Advanced*)
- **Finnish** (Satisfactory)
- Turkish (*Basic*)

REFERENCES

- **Riikka Suovanen**
  Environmental Leader, Paroc Oy
  Riikka.Suovanen@owenscorning.com
- **Okko Niskanen**
  Sr. Engineer R&D, Paroc Oy
  Okko.Niskanen@owenscorning,com

Monitoring and Evaluation Officer

City District Government Lahore | 2021- 2023
Planning and Development Department

- Planned, monitored, and evaluated development projects, ensuring alignment with EPA regulations, Environmental Impact Assessments (EIA), and permitting requirements.
- Contributed to the development of the city's sustainability plans for **waste management** and mobility, including environmentally safe landfill site planning.
- Coordinated public–private initiatives for waste transportation and recycling, supporting process optimization and improved **operational efficiency**.
- Supported the enhancement of process monitoring and reporting systems through integration of **RFID, GIS (ArcGIS)**, and **automated data tools**, enabling improved real-time tracking and analysis of environmental and operational data.
- Contributed to the development of a real-time **digital monitoring platform** for fleet tracking and performance management.

Environmental Coordinator

Integrated Consulting Services Limited | 2020- 2021
Environmental consulting and compliance services

- Delivered consulting support on environmental monitoring, **compliance audits**, and ISO 14001 / ISO 9001 alignment, helping clients meet regulatory and management system requirements.
- Conducted environmental compliance assessments and audits, tracking findings and supporting implementation of corrective actions. Prepared Environmental Impact Assessments (**EIA**), including **wastewater analysis**, environmental risk assessments (**ERA**), and emissions evaluations.
- Supported regulatory permitting processes and advised on compliance strategies aligned with **environmental legislation and standards**. Interpreted environmental data and translated findings into practical, actionable recommendations for operational and compliance improvements.
- Provided **EHS**-related reporting and documentation, supporting safe and compliant project execution